

Ex-4 Dimensionality Reduction using PCA

Dimensionality Reduction using Principal Component Analysis (PCA)

Dataset: Use the Iris Dataset available in Scikit-learn.

The Iris dataset contains four features:

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width

Reduce these four features into two principal components using PCA and visualize the transformed data.

Tasks

1. Load the Iris dataset (It is available in the sci-lit learn library)
2. Display the first five records.
3. Standardize all features.
4. Apply PCA with 2 components.
5. Display the transformed dataset.
6. Find the explained variance ratio.
7. Calculate total variance retained.
8. Plot a scatter graph of PC1 versus PC2.
9. Compare dataset shape before and after PCA.
10. Write your observations.